DECISION TREE

[1]:
```
import pandas as pd
import pickle
import matplotlib.pyplot as plt
```

[2]:
```
dataset = pd.read_csv('Startups_dataset.csv')
dataset
```

[2]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	State	Profit
0	0	165349.30	136897.90	471784.20	New York	192261.93
1	1	162597.80	151377.69	443898.63	California	191792.16
2	2	153441.61	101145.65	407934.64	Florida	191050.49
3	3	144372.51	118671.95	383199.72	New York	182902.09
4	4	142107.44	91391.87	366168.52	Florida	166188.04
5	5	131877.00	99814.81	362861.46	New York	156991.22
6	6	134615.56	147198.97	127716.92	California	156122.61
7	7	130298.23	145530.16	323876.78	Florida	155752.70
8	8	120542.62	148719.05	311613.39	New York	152211.87

[3]:
```
# Apply one-hot encoding
dataset = pd.get_dummies(dataset, columns=['State'], drop_first=True)
dataset
```

[4]:
```
x = dataset[["R&D Spend", "Administration", "Marketing Spend", "State_Florida", "State_New York"]]
y = dataset[["Profit"]]
```

[16]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.20,random_state=43)
```

[17]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

[126]:
```
from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor(criterion='squared_error', splitter='random', max_features = None, random_state=None)
regressor.fit(x_train, y_train)
```

[126]:

DecisionTreeRegressor

▸ Parameters

▸ Fitted attributes

[127]:
```
y_pred = regressor.predict(x_test)
y_pred
```

[127]:
```
array([125370.47,  78240.01, 108734.09, 129917.14, 110352.35,  49490.85,
        78240.01, 155752.7 , 125370.47,  69759.08])
```

```
[128]: from sklearn.metrics import r2_score
       score = r2_score(y_test,y_pred)
       score

[128]: 0.9294150580717183
```

RANDOM FOREST

```
[1]: import pandas as pd
     import pickle
     import matplotlib.pyplot as plt
     from sklearn.model_selection import train_test_split
     from sklearn.preprocessing import StandardScaler
     from sklearn.ensemble import RandomForestRegressor
     from sklearn.metrics import r2_score

[2]: dataset = pd.read_csv('Startups_dataset.csv')
     #dataset

[3]: dataset = pd.get_dummies(dataset, columns=['State'], drop_first=True)
     #dataset

[4]: x = dataset[["R&D Spend", "Administration", "Marketing Spend", "State_Florida", "State_New York"]]
     y = dataset[["Profit"]]

[5]: x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.20,random_state=37)

[6]: sc = StandardScaler()
     x_train=sc.fit_transform(x_train)
     x_test=sc.transform(x_test)
```

```
[7]: regr = RandomForestRegressor(criterion='squared_error', max_depth = None, random_state=2)
     regr.fit(x_train, y_train)

     C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector
     lease change the shape of y to (n_samples,), for example using ravel().
       return fit_method(estimator, *args, **kwargs)

[7]: RandomForestRegressor
     ▸ Parameters
     ▸ Fitted attributes

[8]: y_pred = regr.predict(x_test)
     y_pred

[8]: array([ 57132.4156,  94320.4993, 128095.4114, 185067.1061,  81968.5395,
            148475.9793, 100631.378 , 138872.1447,  99211.3131, 168383.8351])

[9]: score = r2_score(y_test,y_pred)
     score

[9]: 0.9139731737836414
```